



Paper ID: 250693

Printed Page: 1 of 1
Subject Code: RCA1803

Roll No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

MCA-INT
(SEM VIII) THEORY EXAMINATION 2024-25
MOBILE COMPUTING

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****2 x 7 = 14**

a.	What are the various characteristics of mobile computing?
b.	Differentiate the use of HLR and VLR in GSM network.
c.	What is the role of Mobile IP in mobile computing?
d.	What are different wireless multiple access protocols?
e.	What is the general goal of a file system?
f.	What is different fault tolerance issue involved in mobile agent computing?
g.	Write short note on the Global State Routing.

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

a.	What is general packet radio service (GPRS)? Describe its architecture in detail.
b.	What kind of problems may arise if TCP is implemented over wireless networks? Explain
c.	Explain the key challenges in maintaining data consistency during disconnected operations?
d.	Discuss the possible ways for providing security to mobile agents.
e.	Discuss about Adhoc on demand distance vector and its features.

SECTION C**3. Attempt any one part of the following:****7 x 1 = 07**

a.	What is mobile computing? Explain different issues involved in mobile computing.
b.	What is the difference between GSM and GPRS? What are the network elements in GPRS that are different from GSM?

4. Attempt any one part of the following:**7 x 1 = 07**

a.	Define Wireless LAN and give details about the design aspects of IEEE 802.11?
b.	Describe the WAP(Wireless Application Protocol) Application Environment.

5. Attempt any one part of the following:**7 x 1 = 07**

a.	Why is data replication needed in mobile environment? Discuss different replication schemes used for this purpose.
b.	How clustering is done in wireless transmission? Give any suitable example.

6. Attempt any one part of the following:**7 x 1 = 07**

a.	What do you understand by mobile agent? Explain the Application Domains for mobile agent.
b.	Discuss the challenges in transaction processing. What are counter measures to security threat in mobile computing environment?

7. Attempt any one part of the following:**7 x 1 = 07**

a.	How does dynamic source routing handle routing? What is the motivation behind DSR compared to other routing algorithm from fixed network?
b.	Name the main differences between Adhoc network and other network. What advantage do Adhoc Network offer? Explain in detail by giving suitable example.